

Biweekly Report (2026-04-13 ~ 2026-04-26)

work hours: 70h

冯俊财: 20h 路清怡: 15h 张诗蔻: 15h 纪鹏: 20h

I. Phase Objective

This phase focused on the M2 milestone: completing the minimum closed loop of “upload course materials -> build a course knowledge base -> answer questions based on course materials,” while making the system runnable, demonstrable, and measurable.

II. Project Progress

This phase delivered the core M2 workflow: course material upload, asynchronous resource processing, knowledge ingestion, RAG-based Q&A, and offline evaluation. On the backend side, the full pipeline for parsing, chunking, embedding storage, and resource status write-back was completed, with support for PDF, Word, PPT, and Excel materials. Key engineering issues such as worker configuration, embedding dimension mismatch, and embedding batch-size limits were also resolved.

On the intelligent Q&A side, the chat interface was connected to `QAagent`, and the course-aware RAG pipeline was completed with query embedding, chunk embedding, and top-k retrieval. On the frontend side, the course management page and chat page were integrated with backend APIs, enabling course creation, material upload, resource status refresh, and intelligent Q&A directly from the interface. The team also organized course demo materials and question sets, including a structured checklist for the Data Structures course.

At present, the system has completed the core closed loop of “course material upload -> asynchronous processing -> knowledge ingestion -> course Q&A.” For the Data Structures course, an offline evaluation on 25 questions achieved 24 correct answers, corresponding to 96% accuracy, which exceeds the M2 target of 90%.

IV. Issues in This Phase

- The resource processing pipeline encountered issues such as embedding dimension mismatch, embedding batch-size limits, and unstable worker configuration.
- During frontend integration, issues appeared in resource status refresh and in the clarity of page-level error presentation.
- The local development environment on Windows also showed encoding and permission compatibility issues, which required script and configuration adjustments to ensure stable execution.